Scheme of Valuation (MAT-2121)

12 September 2025, (10.45AM-12.15PM)

MCQ

1) 12.5 2) Parabolic 3) Sine terms

4) 0.1389 5) Parabolic 6) -150 7) $O(h^2)$

8) 0 9) $\frac{-4}{n\pi}$ for odd n and 0 for even n 10) -1

Descriptive

11. $N = 6$, $l = 3$ and $\theta = \frac{\pi x}{3}$ ——— (½ M)

$\Sigma y = 42$, $\Sigma y\cos\theta = -8.5$, $\Sigma y\sin\theta = 12.99$ —(2M)

Thus $a_0 = \frac{2.46}{6} = 14$ ——— (½ M)

$$a_n = 2\left(\frac{-8.5}{6}\right) = -2.8333$$
$$b_n = 2\left(\frac{12.990}{6}\right) = 4.3300$$
} — (½ M)

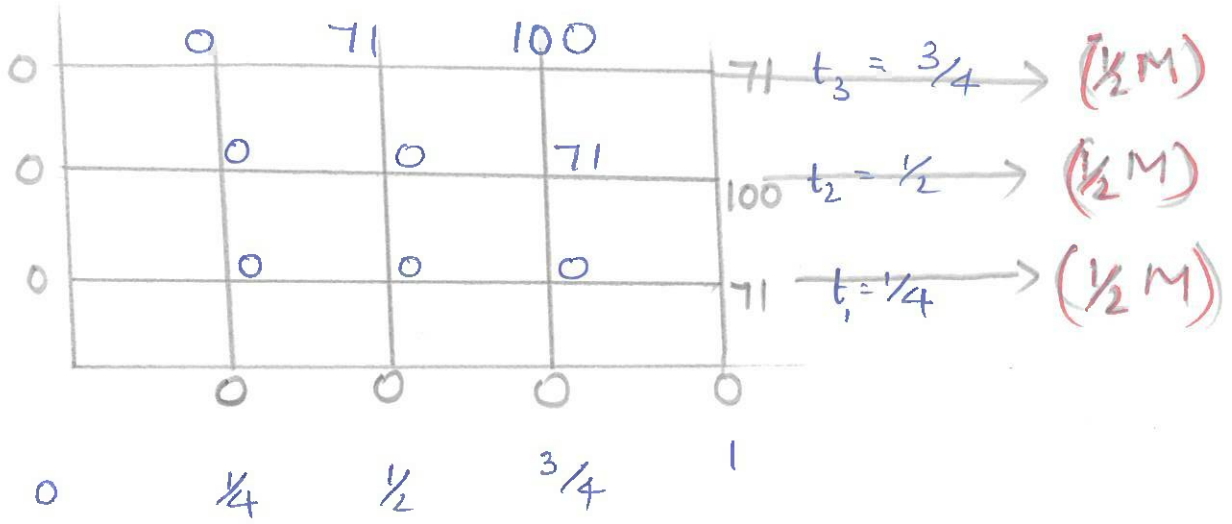
$\therefore y = 7 + (-2.833\cos x + 4.33\sin x) + \ldots$ —(½ M)

12. $c = 1$, $k = \frac{h}{c} = 0.25$

$t_j = jk = j/4$

} → (½ M)

Table with boundary conditions — (1M)



(½M) (½M) (½M)

13. $f(2\pi - x) = x(2\pi - x) = f(x)$

$\therefore f(x)$ is an event function $\Rightarrow b_n = 0$ } — (1M)

$$a_0 = \frac{2}{\pi}\int_0^{\pi} x(2\pi - x)\,dx \Rightarrow a_0 = \frac{4\pi^2}{3} \quad \text{— (½M)}$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} x(2\pi - x)\cos nx\,dx$$

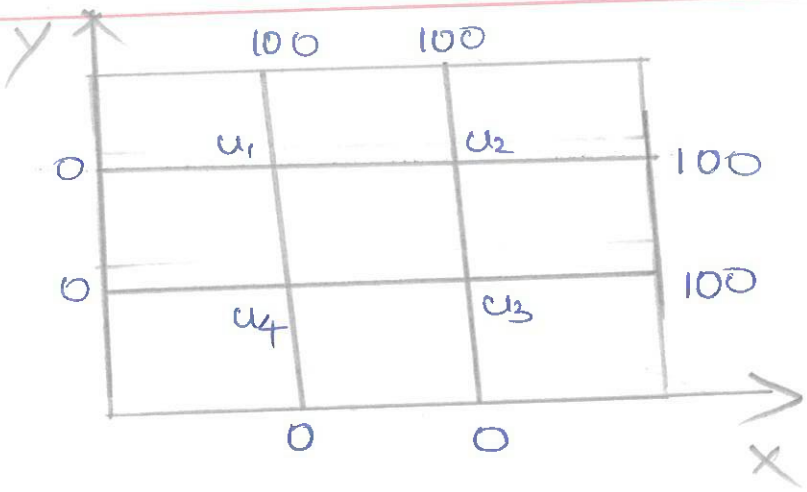
$$= \frac{2}{\pi}\left\{\frac{2\pi(-1)^n}{n^2} - \frac{2\pi}{n^2} - \frac{2\pi(-1)^n}{n^2}\right\} = \frac{-4}{n^2} \quad \text{— (1M)}$$

$$\therefore x(2\pi - x) = \frac{2\pi^2}{3} - 4\sum_{n=1}^{\infty}\frac{1}{n^2}\cos nx$$

For $x = \pi$, we get — (½M)

$$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots$$

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—— (½ M)

By diagoal symmetry $u_1 = u_3$ —— ½ M

$$u_1 = \frac{1}{4}(u_2 + u_4 + 102) \quad \text{——} \ (1)$$

$$u_2 = \frac{1}{4}(u_1 + u_3 + 204) \quad \text{——} \ (2)$$

$$u_4 = \frac{1}{4}(u_1 + u_3 + 1) \quad \text{——} \ (3)$$

} (1½ M)

Solving (1), (2) and (3)

$u_1 = u_3 = 51$, $u_2 = 76$, $u_4 = 26$. —— (½ M)

15 $x_0 = 0$ $x_1 = 1/3$ $x_2 = 2/3$ $x_3 = 1$ —— (½ M)

$y_0 = 0$ $y_1 = ?$ $y_2 = ?$ $y_3 = 0$

For $i = 1$, $-14y_1 + 7y_2 = 4$ —— (1)

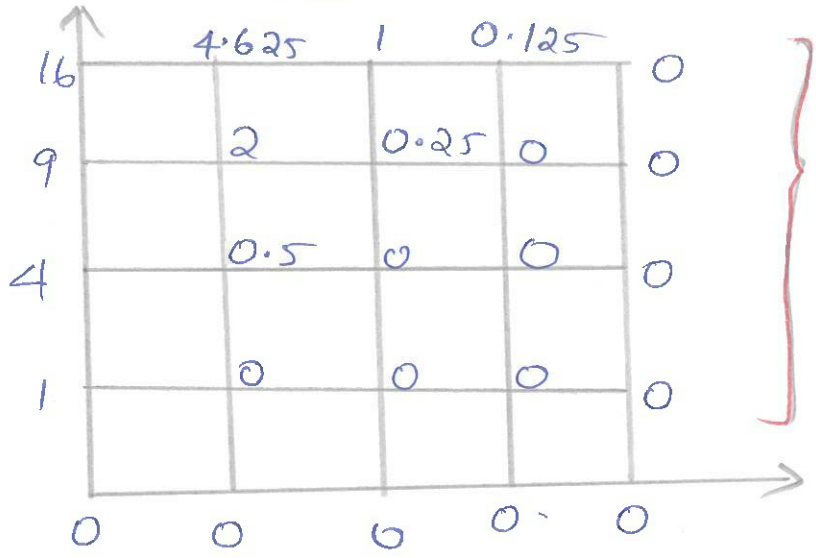
$i = 2$ $13y_1 - 14y_2 = 4$ —— (2)

} —— (2 M)

Solving (1) and (2) $y_1 = -4/5$, $y_2 = \frac{-36}{35}$ —— (½ M)

16.

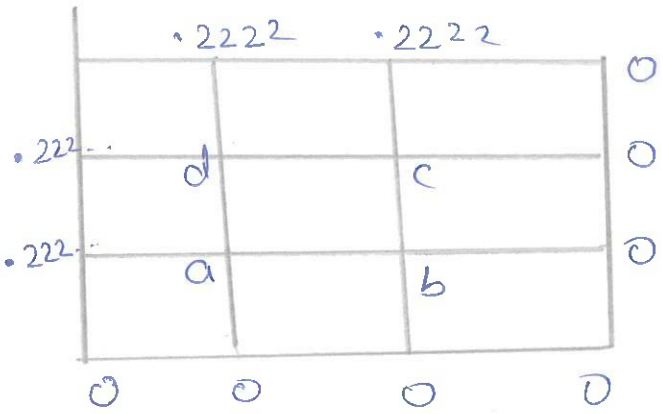
$$k = \frac{\lambda h^2}{c^2} = \frac{\frac{1}{2}\left(\frac{1}{16}\right)}{\frac{1}{32}} = 1.$$ ——— (½ M)



——— (1½ M)

Table with boundary conditions ——— (½ M)

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——— (1 M)

By symmetry $a = c$.

$4a = b + d + 0.2222$

$4b = 2a$

$4d = 2a + 0.4444$

——— (½ M)

Solving $a = 0.1111$, ~~$b = 0.5$~~ $b = 0.0556$ ——— (½ M)

$d = 0.1666$

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$f(x) = x/2$ is an odd function. } — (1M)

$\therefore a_0 = a_n = 0$.

$$b_n = \frac{2}{\pi}\int_0^{\pi} \frac{x}{2}\sin nx\,dx = \frac{(-1)^{n+1}}{n}$$

$$\therefore \frac{x}{2} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}\sin nx. \quad \Big\} \text{— (1M)}$$

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$$a_0 = 2\int_0^1 (x-1)\,dx = -1 \quad \text{— (½M)}$$

$$a_n = 2\int_0^1 (x-1)\cos n\pi x\,dx = \frac{2}{n^2\pi^2}\left[(-1)^n - 1\right] \quad \text{— (1M)}$$

$$\therefore f(x) = \frac{-1}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi^2}\left[(-1)^n - 1\right]\cos n\pi x \quad \text{— (½M)}$$